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Hannes Stärk

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Education

Massachusetts Institute of Technology (MIT), PhD in EECS | Machine Learning — June 2022 – present
- Advised by Prof. Tommi Jaakkola and Prof. Regina Barzilay.

Technical University of Munich (TUM), M.Sc. Informatics | Machine Learning — Oct 2019 – Sept 2021
- Passed with high distinction (1.2); no corrections for Master's thesis.

Bundeswehr University Munich, B.Sc. Informatics | Mathematics Track — Oct 2017 – Sept 2019
- Only student to complete the 3-year curriculum in 2 years.

Teaching Experience

Modeling with Machine Learning: From Algorithms to Applications — MIT — Spring 2025
- Instructor. Designed lectures, problem sets, and exams; held office hours.

Operations Research — Technical University of Munich, Decision Sciences — Apr 2021 – Sept 2021
- Instructor. Taught two recitations per week for ∼40 students; supported online forum.

Deep Learning — Technical University of Munich, CV & AI Niessner Lab — Nov 2020 – Apr 2021
- Instructor. Weekly office hours; created exercises and Jupyter notebooks; answered questions in online forum.

Work Experience

Boltz PBC — Founding Research Scientist (Cambridge, MA) — April 2025 – present

NVIDIA — Research Intern (Santa Clara, CA) — May 2024 – Aug 2024
- Generative models for protein structure conditioned on biologist-specified inputs; published at ICLR 2025. Supervisor: Karsten Kreis.

Valence Discovery — ML Research Intern — Mar 2022 – May 2022

BIB Augsburg gGmbH — Mathematics Instructor (Augsburg, DE) — Feb 2020 – Nov 2021
- 4h/week: teaching linear algebra, analysis, and statistics; online lectures and weekly individual lessons.

Bundeswehr University Munich — Student Assistant (Munich, DE) — Sept 2018 – July 2019
- 10h/week: causal inference and structure learning in Bayesian networks.

Allianz Deutschland AG — Dual Study Program (Munich, DE) — Sept 2017 – Sept 2019
- 38h/week: web development and digital infrastructure in an agile team; technical training in computer science.

Research Dissemination

Starkly Speaking Reading Group — Organizer (virtual) — Aug 2021 – present
- Weekly Zoom discussions with paper authors; ∼75 attendees; Slack community >4,000 researchers.

Molecular Machine Leaning Conference (MIT) — Founder (Cambridge, MA) — Oct 2022 – present
- ∼300-attendee conference in honor of Octavian Ganea.

Learning on Graphs Conference — Co-founder (virtual) — Dec 2022 – Dec 2024
- Reviewing with financial incentives; decentralized meetups; free registration; fully open access.

Invited Talks

Stanford AI+Biomedicine Seminar — Nov 2025

University of California, San Francisco	Nov 2025
Stanford AI+Biomedicine Seminar	Jul 2024
American Chemical Society (Skolnik Award Symposium)	Aug 2023
University of Cambridge — AI Research Seminar	Oct 2021
Technical University of Munich — Two guest lectures on protein prediction	Nov 2021

Publications

Note: * denotes equal first authorship; † denotes corresponding authorship.

Highlighted Preprints

BoltzGen: Toward Universal Binder Design 2025
H. Stärk†, F. Faltings, M. G. Choi, Y. Xie, E. Hur, T. O'Donnell, A. Bushuiev, T. Uçar, S. Passaro, W. Mao, M. Reveiz, R. Bushuiev, T. Pluskal, J. Sivic, K. Kreis, A. Vahdat, S. Ray, J. T. Goldstein, A. Savinov, J. A. Hambalek, A. Gupta, D. A. Taquiri-Diaz, Y. Zhang, A. K. Hatstat, A. Arada, N. H. Kim, E. Tackie-Yarboi, D. Boselli, L. Schnaider, C. C. Liu, G.-W. Li, D. Hnisz, D. M. Sabatini, W. F. DeGrado, J. Wohlwend, G. Corso, R. Barzilay, T. Jaakkola. *Under review at Nature.*

Boltz-2: Towards accurate and efficient binding affinity prediction 2025
S. Passaro*, G. Corso*, J. Wohlwend*, M. Reveiz*, S. Thaler*, V. R. Somnath, N. Getz, T. Portnoi, J. Roy, **H. Stärk**, D. Kwabi-Addo, D. Beaini, T. Jaakkola, R. Barzilay. *Under review at Nature.*

Conference Papers (NeurIPS / ICML / ICLR)

ProtComposer: Compositional Protein Structure Generation with 3D Ellipsoids 2025
H. Stark*, B. Jing*, T. Geffner, J. Yim, T. Jaakkola, A. Vahdat, K. Kreis. *ICLR Oral. Among 0.7% best scores*.

Think while you generate: Discrete diffusion with planned denoising 2025
S. Liu, J. Nam, A. Campbell, **H. Stärk**, Y. Xu, T. Jaakkola, R. Gómez-Bombarelli. *ICLR*.

ET-Flow: Equivariant Flow-Matching for Molecular Conformer Generation 2025
M. Hassan, N. Shenoy, J. Lee, **H. Stärk**, S. Thaler, D. Beaini. *NeurIPS*.

Dirichlet Flow Matching with Applications to DNA Sequence Design 2024
H. Stärk*, B. Jing*, C. Wang, G. Corso, B. Berger, R. Barzilay, T. Jaakkola. *ICML*

Harmonic Self-Conditioned Flow Matching for Multi-Ligand Docking and Binding Site Design 2024
H. Stärk*, B. Jing*, R. Barzilay, T. Jaakkola. *ICML*.

Generative Modeling of Molecular Dynamics Trajectories 2024
B. Jing*, **H. Stärk***, T. Jaakkola, B. Berger. *NeurIPS*.

DiffDock: Diffusion Steps, Twists, and Turns for Molecular Docking 2023
G. Corso*, **H. Stärk***, B. Jing*, R. Barzilay, T. Jaakkola. *ICLR*.

EquiBind: Geometric Deep Learning for Drug Binding Structure Prediction 2022
H. Stärk*, O. E. Ganea*, L. Pattanaik, R. Barzilay, T. Jaakkola. *ICML*.

3D Infomax improves GNNs for Molecular Property Prediction 2022
H. Stärk, D. Beaini, G. Corso, P. Tossou, C. Dallago, S. Günnemann, P. Liò *ICML*.

Journal Articles

Protein FID: Improved Evaluation of Protein Structure Generative Models 2025
F. Faltings, **H. Stärk**, T. Jaakkola, R. Barzilay. *Bioinformatics*.

Evolutionary design of conductive pathways using a generative autoencoder 2025
M. C. Ignuta-Ciuncanu, **H. Stärk**, R. F. Martinez-Botas. *International Communications in Heat and Mass Transfer* 166:109098.

Graph neural networks 2024
G. Corso*, **H. Stark***, S. Jegelka, T. Jaakkola, R. Barzilay. *Nature Reviews Methods Primers* 4(1):17.

Diffusion models in protein structure and docking 2024
J. Yim, **H. Stärk**, G. Corso, B. Jing, R. Barzilay, T. S. Jaakkola. *WIREs Comput. Mol. Science* 14(2):e1711.

Blind protein-ligand docking with diffusion-based deep generative models 2023
G. Corso, **H. Stark**, R. Barzilay, T. Jaakkola. *Biophysical Journal* 122(3):143a.

Benchmarking AlphaFold-enabled molecular docking predictions for antibiotic discovery 2022
F. Wong, A. Krishnan, E. J. Zheng, **H. Stärk**, A. L. Manson, A. M. Earl, T. Jaakkola, J. J. Collins. *Molecular Systems Biology* 18(9):e11081.

Equivariant 3D-Conditional Diffusion Models for Molecular Linker Design 2022
I. Igashov, **H. Stärk**, C. Vignac, V. G. Satorras, P. Frossard, M. Welling, M. Bronstein, B. Correia. *Nature Machine Intelligence* 6(4):417–427.

Light Attention Predicts Protein Location from the Language of Life 2021
H. Stärk, C. Dallago, M. Heinzinger, B. Rost. *Bioinformatics Advances*.

Preprints

ProxelGen: Generating Proteins as 3D Densities 2025
F. Faltings*, **H. Stärk***, R. Barzilay, T. Jaakkola. *Preprint.*.

Training on test proteins improves fitness, structure, and function prediction 2024
A. Bushuiev, R. Bushuiev, N. Zadorozhny, R. Samusevich, **H. Stärk**, J. Sedlar, T. Pluskal, J. Sivic. *Preprint.*

CodonMPNN for Organism Specific and Codon Optimal Inverse Folding 2024
H. Stark, U. Padia, J. Balla, C. Diao, G. Church. *Preprint.*.

PINDER: The protein interaction dataset and evaluation resource 2024
D. Kovtun, M. Akdel, A. Goncearenco, G. Zhou, G. Holt, D. Baugher, D. Lin, Y. Adeshina, T. Castiglione, X. Wang, C. Marquet, M. McPartlon, T. Geffner, E. Rossi, G. Corso, **H. Stärk**, Z. Carpenter, E. Kucukbenli, M. Bronstein, L. Naef. *Preprint*.

PLINDER: The protein-ligand interactions dataset and evaluation resource 2024
J. Durairaj, Y. Adeshina, Z. Cao, X. Zhang, V. Oleinikovas, T. Duignan, Z. McClure, X. Robin, G. Studer, D. Kovtun, E. Rossi, G. Zhou, S. Veccham, C. Isert, Y. Peng, P. Sundareson, M. Akdel, G. Corso, **H. Stärk**, G. Tauriello, Z. Carpenter, M. Bronstein, E. Kucukbenli, T. Schwede, L. Naef. *Preprint*.

FreeFlow: Latent Flow Matching for Free Energy Difference Estimation 2024
E. Erdogan*, R. Ralev*, M. Rebensburg*, C. Marquet, L. Klein, **H. Stark**. *Preprint*.

Transition Path Sampling with Boltzmann Generator-based MCMC Moves 2023
M. Plainer*, **H. Stärk***, C. Bunne, S. Günnemann. *Preprint.*

DiffDock-Pocket: Diffusion for Pocket-Level Docking with Sidechain Flexibility 2023
M. Plainer*, M. Toth*, S. Dobers*, **H. Stark**, G. Corso, C. Marquet, R. Barzilay. *Preprint*.

DiffDock-PP: Rigid Protein-Protein Docking with Diffusion Models 2023
M. A. Ketata*, C. Laue*, R. Mammadov*, **H. Stärk**, M. Wu, G. Corso, C. Marquet, *et al. Preprint.*

Latent Space Simulator for Unveiling Molecular Free Energy Landscapes and Predicting Transition Dynamics 2023
S. Dobers, **H. Stark**, X. Fu, D. Beaini, S. Günnemann. *Preprint.*

Generalized Laplacian Positional Encoding for Graph Representation Learning 2022
S. Maskey, A. Parviz, M. Thiessen, **H. Stärk**, Y. Sadikaj, H. Maron. *Preprint.*

Graph Anisotropic Diffusion 2022
A. A. A. Elhag, G. Corso, **H. Stärk**, M. M. Bronstein. *Preprint.*

Jointly Learnable Data Augmentations for Self-Supervised GNNs 2021
Z. T. Kefato, S. Girdzijauskas, **H. Stärk**. *Preprint.*